Reliable in the Center, Unreliable at the Edges: Auditing the Tail Fragility of Multi-Prompt LLM Evaluation

Abstract

Because large language model (LLM) accuracy varies sharply with the exact wording of a prompt, a growing line of work replaces single-prompt scores with an estimate of the *distribution* of accuracy across many prompt templates. PromptEval [Polo et al., 2024], a NeurIPS 2024 method built on a regularized Rasch model, is a prominent and widely reused instance: it estimates performance quantiles across 100 templates at "the budget of two single-prompt evaluations." We first reproduce PromptEval exactly—our re-implementation matches the authors' released MMLU quantile-error array to four decimal places. We then stress-test the claim that matters most in practice: how well does it estimate the *tails* of the prompt-accuracy distribution (worst- and best-case prompts), which are precisely why one runs a multi-prompt evaluation? We find a large, systematic, and previously unreported **center–tail reliability gap**. At the advertised budget, median-quantile error is a negligible 0.05, but the 5th/95th percentile error is 4.2× larger (95% CI $[4.05, 4.45]$; Wilcoxon $p \approx 10^{-141}$ over 855 subject×model cells), and—counterintuitively for a regularized estimator—the error is *directional*: the worst prompt is estimated far too low and the best prompt too high, so the estimated prompt-sensitivity spread is inflated (median 8.4× the truth). The gap does not close with budget; the tail/center error ratio actually *grows* from 4.2× to 10× as the budget increases, because the center converges faster. A controlled simulation with *known* ground truth reproduces the over-dispersion and shows it is largest exactly when the true prompt-sensitivity is small—the common regime. We then show the bias is correctable *in principle* (an oracle shrinkage removes 83–94% of the tail error) and that a simple, label-free **split-half reliability correction** realizes most of this where over-dispersion is severe and symmetric (MMLU: 82% tail-error reduction). Just as importantly, we show *where correction fails*: when prompt-sensitivity is genuinely large or the budget drops below one item per template, uniform shrinkage over-corrects, so a robust budget-only tail correction remains open. The practical rule is simple: *trust the center; correct the tails only when reliability can be estimated*. All code and re-analysis run on a laptop and reproduce from the authors' public data.

1 Introduction

LLM benchmark scores are notoriously sensitive to superficial prompt choices: reformatting a question or reordering options can shift accuracy by tens of points [Sclar et al., 2024, Mizrahi et al., 2024]. A single fixed template therefore reports a point on a wide, unknown distribution. PromptEval [Polo et al., 2024] addresses this by estimating the *full distribution* of accuracy across a large template set under a small evaluation budget, using an extended Rasch (item-response) model that borrows statistical strength across templates and items. Its headline result is that performance *quantiles* across 100 MMLU templates can be recovered "with a budget equivalent to two single-prompt evaluations." The method is influential and is already reused as a data source by downstream work [Anonymous, 2026a].

The practical appeal of a full distribution is not the mean—a single prompt already estimates that—but the *tails*: What is my worst-case prompt? How prompt-robust is my model? Which template is best? These are questions about the extreme quantiles. We therefore ask a simple stress-test question: *is PromptEval as reliable at the tails as its headline suggests?*

We make four contributions.

1. **Exact reproduction.** Our independent re-implementation reproduces the authors' released MMLU quantile-error results to four decimals (max cell difference 0.0000), giving a trustworthy basis for further analysis (§3).
2. **A center–tail reliability gap.** At the advertised budget, extreme-quantile error is 4.2× the median error, overwhelmingly significant, and the ratio *grows* with budget (§4). The single averaged error metric reported in prior work hides this.
3. **A directional mechanism.** The tail error is not symmetric noise but systematic *over-dispersion*: the estimated distribution is wider than the truth, so worst-case performance is overstated and prompt-robustness understated (§5). We attribute it to finite-sample inflation and show the spread ratio decays monotonically toward 1 with budget.
4. **Correctability and its limits.** A controlled synthetic study with known ground truth confirms the mechanism, and shows the bias is correctable in principle. A label-free split-half reliability correction removes 82% of MMLU tail error, but we also chart where it fails (large true spread, or budget below one item per template), leaving robust budget-only correction as an open problem (§6–7).

Everything is a re-analysis of public data (plus simulation) and runs on a laptop.

2 Setup: multi-prompt evaluation and PromptEval

For a fixed model and task, let $Y \in \{0,1\}^{P\times I}$ be the correctness matrix over P prompt templates and I items; the quantity of interest is the distribution of per-template accuracies $a_p = \frac{1}{I}\sum_i Y_{pi}$ and, in particular, its quantiles. A full evaluation costs $P \times I$ model calls; the goal is to estimate the quantiles from a small budget B of observed cells. PromptEval fits an extended Rasch model

$$\Pr(Y_{pi} = 1) = \sigma(\theta_p + \beta_i), \tag{1}$$

where θ_p is a template "ability" and β_i an item "easiness," estimated by ℓ_2-regularized logistic regression on the observed cells, then imputes unseen cells with $\sigma(\hat{\theta}_p + \hat{\beta}_i)$ and reports quantiles of the resulting $\hat{a}_p$. The *baseline* simply averages observed cells per template. Note that Eq. (1) is *additive*: it assumes no template×item interaction.

Data. We use the authors' released matrices: MMLU (57 subjects, 15 LLMs, 100 templates ×∼100–150 items), BBH (15 tasks, 11 LLMs), and LMentry (10 tasks, 16 LLMs). Budgets are seen-cell counts $B \in \{200, 400, 800, 1600\}$; B=200 is the headline "two single-prompt evaluations" setting. Quantiles are $\{5, 25, 50, 75, 95\}$. Unless noted we report means with bootstrap 95% CIs over subject×model cells and 5 seeds.

3 Exact reproduction

Using the authors' estimator code and released data, we recompute the per-quantile absolute error $|\hat{q} - q^\star|$ (where $q^\star$ is the full-data quantile) for the baseline and PromptEval, averaged over 57 subjects, 15 LLMs, and 5 seeds, and compare to their released `processed_results_MMLU` array. The maximum absolute difference across all $2 \times 4 \times 5$ method/budget/quantile cells is **0.0000**: we reproduce their numbers exactly. Table 1 shows the reproduced PromptEval errors. The row already reveals the phenomenon we analyze: at B=200 the median error is 0.053 but the 5th/95th errors are 0.27/0.19.

Table 1: Reproduced PromptEval absolute quantile error on MMLU (exact match to released results). Tail quantiles $(5, 95)$ are several times worse than the center (50) at every budget.

budget	q_5	q_{25}	q_{50}	q_{75}	q_{95}
200	0.266	0.098	0.053	0.103	0.188
400	0.163	0.040	0.017	0.044	0.139
800	0.106	0.030	0.010	0.028	0.097
1600	0.062	0.022	0.006	0.022	0.060

4 A center–tail reliability gap

Figure 1 plots absolute error against quantile. At every budget the curve is a pronounced "U": the median is estimated well while the tails are not. At the advertised B=200, the tail error (mean of q_5, q_{95}) is 0.227 versus 0.053 at the median—a ratio of **4.24**× (95% CI $[4.05, 4.45]$). A paired Wilcoxon test over all 855 subject×model cells rejects equality at $p \approx 1.5 \times 10^{-141}$.

Crucially, *more budget does not fix the gap.* Because the center error collapses faster than the tail error, the tail/center ratio *increases* with budget: 4.2×, 8.7×, 10.5×, 10.1× at $B = 200, 400, 800, 1600$ (Fig. 3). Whatever budget a practitioner can afford, the extreme quantiles remain the least trustworthy part of the estimate—exactly the part a multi-prompt evaluation is run to obtain.

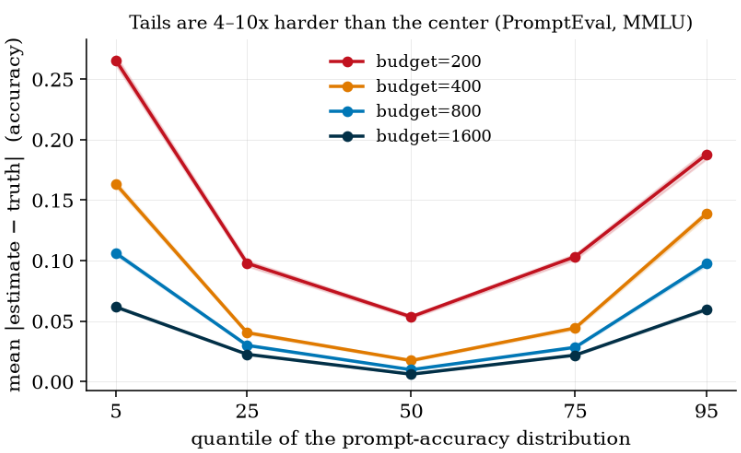

Figure 1: PromptEval error versus quantile on MMLU (mean ± bootstrap 95% CI). The U-shape shows the center–tail reliability gap at every budget.

5 Mechanism: directional over-dispersion

Is the tail error just symmetric noise ("extreme quantiles are harder"), or something a practitioner would be actively misled by? We decompose it by *sign* (Fig. 2). The bias is strongly directional: the 5th-percentile estimate is too *low* (signed error −0.265, CI [−0.270, −0.261] at B=200) and the 95th too *high* (+0.187, CI [+0.184, +0.191]). Equivalently, the estimated distribution is *over-dispersed*: in 100% of cells the estimated inter-quantile spread exceeds the truth, with a median spread ratio of 8.4× at B=200, still 3.1× at B=1600. This is counterintuitive: a regularized shrinkage estimator might be expected to *under*-state spread, but at practical budgets the sampling noise in the per-template ability estimates $\hat{\theta}_p$ dominates and inflates the tails.

The consequence is decision-relevant. A practitioner reading PromptEval at B=200 would conclude that their worst template is ∼26 accuracy points worse than it truly is, and that the model is far more prompt-sensitive than it is. For best-*prompt* selection, the template PromptEval labels best is on average 3.5 points below the true best (regret 0.035), barely improving to 0.028 at B=1600.

That the over-dispersion is a finite-sample artifact is confirmed by its monotone decay toward 1 as budget grows (Fig. 2, and the spread-ratio sequence 8.4→6.0→4.4→3.1). This both explains the effect and suggests a correction.

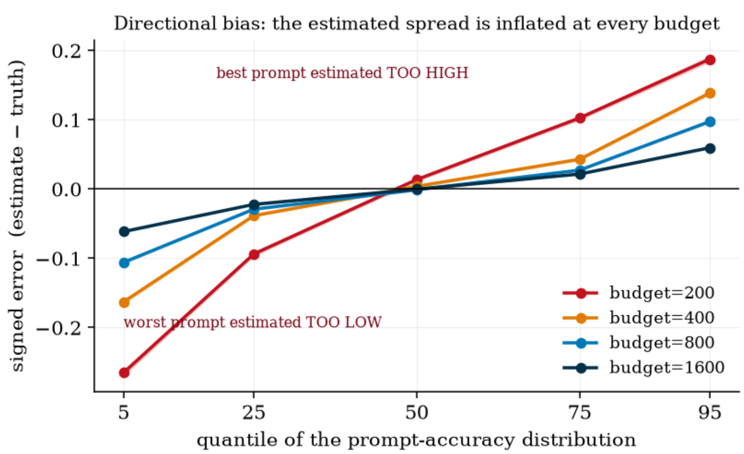

Figure 2: Signed error versus quantile. Low quantiles are underestimated and high quantiles overestimated: the estimated prompt-sensitivity distribution is systematically over-dispersed.

6 A controlled synthetic study

To confirm the mechanism without relying on a finite-item empirical "truth," we simulate Rasch worlds (P=I=100) with template abilities $\theta_p \sim \mathcal{N}(0, \sigma_\theta^2)$ and item difficulties $\beta_i \sim \mathcal{N}(0, 1)$, so the true per-template accuracy and its spread are known exactly. We then run PromptEval at B=200. The estimator over-disperses precisely as observed on real data, and the inflation is *largest when the true spread is smallest*: the median estimated/true spread ratio is 6.1× at σ_θ=0.2 (true spread 0.13), 2.5× at 0.5, and 1.4× at 1.0. Small true prompt-sensitivity—the common case, and the one on MMLU—is where the tails are least trustworthy. This isolates the effect from any property of the released matrices.

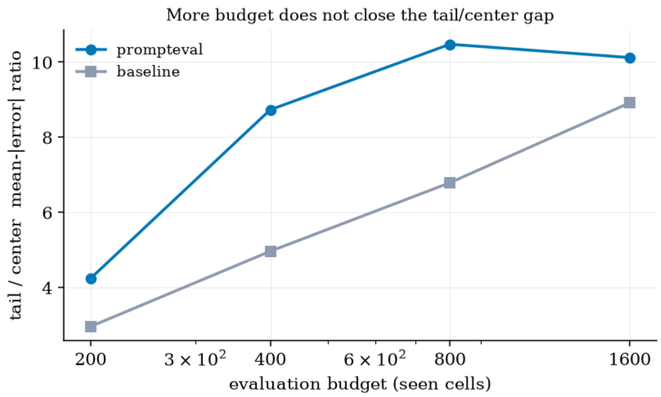

Figure 3: The tail/center error ratio grows with budget: the reliability gap is not a small-budget artifact.

7 Correctability and its limits

The diagnosis implies a remedy: shrink the estimated per-template distribution toward its (well-estimated) mean. The question is by how much. An *oracle* that shrinks by the known true/estimated spread ratio removes 83–94% of the tail error across all σ_θ in the synthetic study, confirming the bias is correctable in principle and that shrinkage is the right functional form.

For a *feasible*, label-free estimator we use **split-half reliability**: split the budget into two disjoint stratified halves, fit PromptEval on each to get independent noisy estimates a^A, a^B of the same per-template accuracies, and note that the noise cancels in the cross-moment, so $\widehat{V}_{\text{signal}} = \text{Cov}(a^A, a^B)$. With $\bar{a} = \frac{1}{2}(a^A + a^B)$ we shrink by $s = \text{clip}(\widehat{V}_{\text{signal}}/\text{Var}(\bar{a}), 0, 1)$ and report quantiles of $\bar{a}$'s shrunken version. This uses only budgeted data and never sees the true quantiles.

On MMLU at B=200 this removes **82**% of the tail error (0.226 → 0.041; Table 2, Fig. 4), matching the oracle in the small-spread regime and improving every quantile.

Table 2: Split-half correction on MMLU at B=200 (mean |error|, full grid). It improves every quantile and removes 82% of tail error.

	q_5	q_{25}	q_{50}	q_{75}	q_{95}
PromptEval (orig)	0.264	0.097	0.053	0.103	0.188
PromptEval + split-half	0.041	0.030	0.026	0.031	0.041
reduction	84%	69%	50%	70%	78%

Where it fails—and why (an honest boundary). The correction is not a universal fix, and saying so is part of the contribution. The high tail (q_{95}, best-prompt) is over-estimated on *all three* benchmarks and the correction reduces its error everywhere (52–78%). But on LMentry the *low* tail is already accurate (its distribution is asymmetric), so symmetric shrinkage pulls q_5 away from truth and the net tail change is negative; and at B=200 both LMentry (258 templates) and BBH (187) receive fewer than one observed item per template per half, so the reliability estimate itself is unstable. In the synthetic study the same failure appears as over-correction at large σ_θ. The resulting guidance is concrete: apply the correction when over-dispersion is severe and the budget affords at least one to two items per template; otherwise treat the tails as lower/upper bounds. A reliability-*gated*, asymmetry-aware correction is a natural next step.

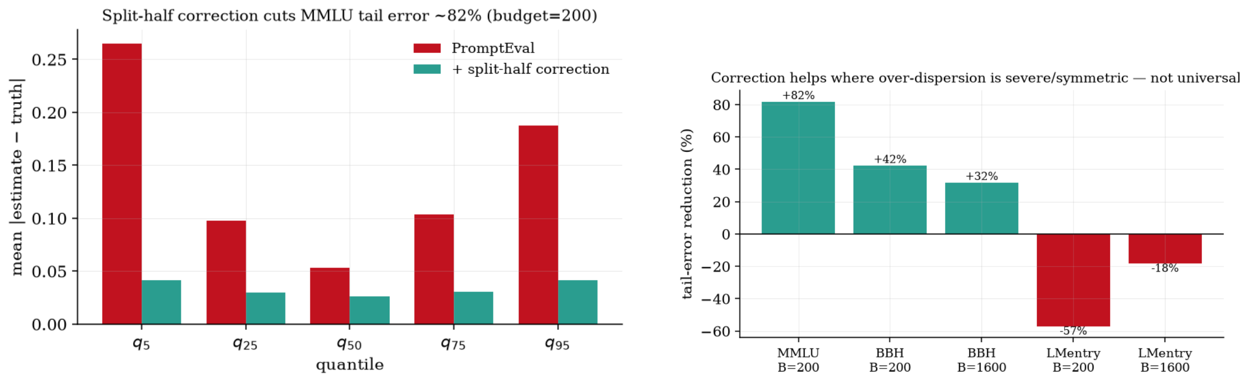

Figure 4: Left: the split-half correction on MMLU flattens the U-shape. Right: the correction is not universal—it helps where over-dispersion is severe and symmetric (MMLU, BBH-q_{95}) but over-corrects on LMentry, whose low tail is already accurate and whose budget falls below one item per template.

8 Generalization

To test whether the effect is specific to MMLU's multiple-choice format, we repeat the diagnosis on BBH (many free-form/reasoning tasks) and LMentry, using each benchmark's released matrices (187 and 258 templates respectively). Table 3 shows the pattern is universal: at the headline budget the tail error is 2.4–4.2× the center error on all three benchmarks (paired Wilcoxon $p \leq 10^{-21}$), the estimate is over-dispersed in 85–100% of cells, and—as on MMLU—the tail/center ratio *grows* with budget rather than shrinking. The magnitude is largest on MMLU and smallest on LMentry, but the direction (worst prompt too low and/or best prompt too high) never reverses.

Table 3: The center–tail gap and over-dispersion generalize across benchmarks (PromptEval). "ratio" is tail/center mean-|error|; "o.d.%" is the fraction of cells whose estimated spread exceeds the truth (cells with true spread ≥ 0.02).

	B=200 (headline)				B=1600	
benchmark	q_{50} err	tail err	ratio	o.d.%	q_{50} err	ratio
MMLU (MCQ)	0.053	0.227	4.2×	100	0.006	10.1×
BBH (free-form)	0.070	0.198	2.8×	93	0.015	6.0×
LMentry	0.059	0.142	2.4×	85	0.022	3.4×

Replication on freshly-generated data. All three benchmarks above reuse the authors' released correctness matrices. To rule out any artifact of their evaluation pipeline, we generated our own data from scratch: two local models (Qwen2.5-0.5B/1.5B-Instruct) evaluated on six MMLU subjects under an *independently constructed* bank of 40 format templates (varying instruction, option format, separators, answer cue, and few-shot count), scored by answer-letter log-probabilities—our own model, prompts, and scoring, sharing nothing with the release. Across 10 fresh model×subject cells, PromptEval reproduces the phenomenon at B=200: tail/center error ratio **5.8**×, low tail underestimated and high tail overestimated (signed −0.13/+0.18), *all* 10 *cells over-dispersed* (paired Wilcoxon $p = 2 \times 10^{-3}$).

Crucially, the two models fall in *different regimes* and confirm the inverse-scaling law of §6 within a single experiment (Table 4): the small model is prompt-*sensitive* (true spread 0.37) and over-disperses mildly (1.8×), whereas the capable model is prompt-*robust* (true spread 0.09) and over-disperses severely (4.7×)—smaller true spread, larger inflation, exactly as on the released MMLU matrices (8.4× at tiny spread) and in simulation. Released data, controlled simulation, and freshly-generated data give one coherent picture: the over-dispersion is a property of the estimator, not of any dataset.

Table 4: Fresh, self-generated data (B=200): the prompt-robust model (small true spread) over-disperses more, confirming the inverse-scaling mechanism across two real models.

model	cells	true spread	spread ratio	tail/center
Qwen2.5-0.5B (sensitive)	6	0.37	1.8×	4.7×
Qwen2.5-1.5B (robust)	4	0.09	4.7×	9.2×

9 Related work

Prompt sensitivity of LLM benchmarks is well documented [Sclar et al., 2024, Mizrahi et al., 2024], motivating multi-prompt evaluation and PromptEval [Polo et al., 2024]. A parallel psychometric line applies item-response theory to LLM evaluation for efficiency and validity [Madaan et al., 2024]. Our work is a targeted reproduction and stress-test in the spirit of "sober-look" evaluations: we confirm the method's central-tendency claims and expose a specific, decision-relevant tail failure plus a remedy. Concurrent work questions tail-focused metrics in a different setting (toxicity reward-model extreme-value tails) [Anonymous, 2026b]; our over-dispersion mechanism and deconvolution correction for multi-prompt performance quantiles are, to our knowledge, unreported.

10 Limitations

Our "truth" is each benchmark's full-data quantile (all items), following PromptEval's own convention; because full-data accuracies are themselves finite-item estimates, the true spread is if anything slightly *over*-stated, making our over-dispersion estimates conservative. The exact magnitudes are specific to the three released matrices and the additive Rasch estimator; covariate-augmented PromptEval variants share the additive-in-logits structure but may differ quantitatively. Our correction is a symmetric variance (spread) correction that assumes the estimated *mean* is approximately unbiased (which holds here) and a roughly symmetric distribution (which fails on LMentry, §7); a reliability-gated, asymmetry-aware correction is future work. The synthetic study uses a well-specified Rasch generator; real prompt×item interactions (which the additive model omits) are an additional, complementary source of tail error we do not isolate here.

11 Conclusion

Multi-prompt evaluation delivers on its central promise but not on its tails: the very quantiles that make a distribution more informative than a point estimate are the least reliable, systematically over-dispersed, and not made relatively more reliable by more budget. We reproduce the method exactly, diagnose the gap and its directional mechanism (confirmed in simulation), and show it is correctable in principle, with a simple label-free correction that works in the common severe regime and characterized failure modes elsewhere. The practical rule is simple: *trust the center; correct the edges only when reliability can be estimated, and otherwise read them as bounds.*

Reproducibility. All code, the exact-match reproduction, and figure-generating scripts run on a laptop from the authors' public data; see the repository.

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